

Cybersecurity Incident Report

Section 1: Identify the type of attack that may have caused this network interruption	
One potential explanation for the website's connection timeout error message is:	A TCP SYN Flood attack.
The logs show that:	The logs show a large number of TCP SYN packets that do not complete the TCP three-way handshake. The server waits for the final ACK packet, creating many half-open TCP connections.
This event could be:	A TCP SYN Flood attack targeting the web server in an attempt to overwhelm its resources.
Section 2: Explain how the attack is causing the website to malfunction	
1.	SYN – The client sends a SYN packet to the server to request a new TCP connection.
2.	SYN-ACK – The server acknowledges the client's request and responds with a SYN-ACK packet.
3.	ACK – The client acknowledges the server's response, and the TCP connection is established.
Explain what happens when a malicious actor sends a large number of SYN packets all at once:	The server waits for ACK packets that never arrive, creating many half-open TCP connections and exhausting server resources.
Explain what the logs indicate and how that affects the server:	The logs indicate a large number of incomplete TCP connections. The server becomes overwhelmed and cannot respond to legitimate users.